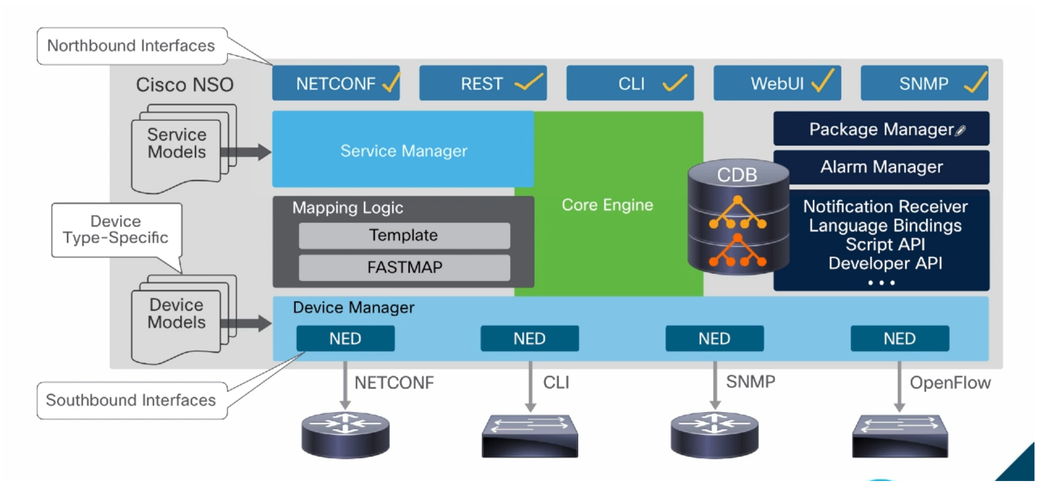
Northbound(User/Application → NSO) *is the interface through which* ***users or applications interact with NSO*** *by sending high-level service requests.*
Southbound(NSO → Devices) *is the interface through which* ***NSO communicates with network*** *devices to apply configurations.*

1. Cisco NSO Service manager receives high-level service requests through northbound interfaces (like Netconf,Restconf,cli), stores them in the candidate configuration,
2. and uses FASTMAP to compare the requested state with the existing state in the CDB (diff calculation).
3. It then applies mapping logic(vendor selection, template selection, and template execution) to generate device-specific configurations.
4. These configurations are passed to the Device Manager and processed by NEDs, which translate them into appropriate protocols (CLI, NETCONF, or RESTCONF),
5. and are then applied to network devices via southbound interfaces. Finally, the CDB is updated during the commit phase.



(Northbound → Core → Southbound)

✅ ✅ ✅ **Cisco NSO End-to-End Orchestration Flow**

🔷 **1. Northbound Layer (Input to NSO)**

✅ **Service Request Source**

- CLI / Web UI (Manual)
- NETCONF / RESTCONF APIs
- SNMP (Monitoring only)

✅ **Purpose**

👉 Users/applications send **high-level service requests (Intent)**

✅ **Example Input**

Create VPN / VLAN / Internet Service

customer = A
bandwidth = 100M
device = R1

```
{
 "name": "A",
 "device": "R1",
 "bandwidth": "100M"
}
```

🔷 2. Service Model (YANG)

✅ Responsibilities

Defines service structure (what inputs are required)
Validates input data (type, format, mandatory fields)
Ensures referenced values exist (e.g., device in CDB)
Links to mapping logic using servicepoint

```
list simple-service {
 key name;
 leaf device {
  type leafref {
   path "/ncs:devices/ncs:device/ncs:name";
  }
 }
 leaf bandwidth {
  type string;
 }
}
```

✅ Flow

Northbound Request
↓
YANG Model Validatio
↓
Valid Input Passed to Service Manager

🔷 2. Service Manager (Entry Point in NSO)

✅ Responsibilities

- Accept request
- Validate input
- Initiate service execution

✅ Flow

Northbound Request

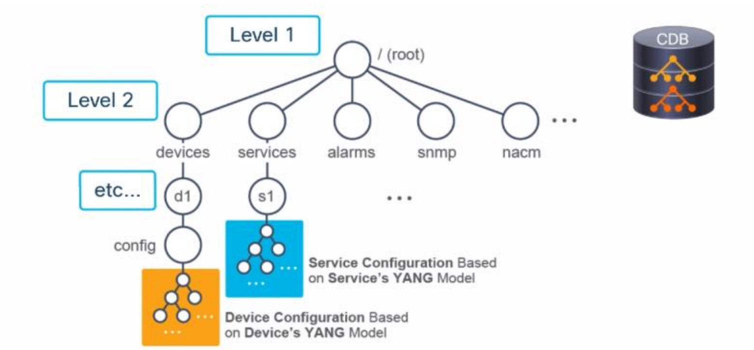
↓

Service Manager

🔷 **3. Configuration Database (CDB)**

✅ **Responsibilities**

- Store:
  - Service data (high-level)
  - Device configurations
  - NSO system state



✅ **Flow**

Service Manager

↓

Store service in Candidate (NOT final CDB yet)

🔷 **4. FASTMAP Engine (Core Processing)**

✅ **Triggered When:**

- Create
- Update
- Delete service

✅ **Responsibilities**

1. Compare:
2. Existing state (CDB) vs Requested state (ie., From Candidate storage)
3. Calculate **DIFF**
4. Decide what to:
   - Create ✅
   - Modify ✅
   - Delete ✅

✅ **Flow**

CDB

↓

FASTMAP Start

↓

State Comparison (Old vs New)

↓

DIFF Calculation

🔷 **5. Mapping Logic Layer (Decision Layer)**

✅ **Purpose**

👉 Convert **service intent → device configuration**

✅ **Responsibilities**

- Vendor selection
- Template selection
- Dynamic logic execution

✅ **Example Logic**

IF vendor == Cisco → cisco-template.xml

IF vendor == Huawei → huawei-template.xml

IF vendor == Nokia → nokia-template.xml

✅ **Implementation**

- YANG Model → Service definition
- Templates (XML/CLI) → Config generation
- Python/Java → Dynamic logic

✅ **Flow**

FASTMAP

↓

Mapping Logic

↓

Vendor selection + Template execution

🔷 **6. Template Execution**

✅ **Responsibility**

- Generate device-specific configuration

✅ **Output Example**

vlan 100
name SALES
interface trunk
 switchport trunk allowed vlan add 100

✅ **Flow**

Mapping Logic
↓
Templates executed
↓
Device config generated

🔷 **7. Device Manager**

✅ **Responsibilities**

- Manage:
  - Device connectivity
  - Authentication
  - Bulk operations

✅ **Flow**

Generated Config
↓
Device Manager

🔷 **8. NED (Network Element Driver)**

✅ **MOST IMPORTANT COMPONENT**

✅ **Responsibilities**

- Translate config into protocol:
  - CLI
  - NETCONF
  - RESTCONF

✅ **Example**

NSO model → CLI command

✅ **Flow**

Device Manager
↓
NED (Protocol conversion)

🔷 **9. Southbound Interface (Execution Layer)**

✅ **Protocols Used**

- CLI (SSH/Telnet)
- NETCONF
- RESTCONF
- SNMP (monitoring)

✅ **Purpose**

👉 Send configuration to actual devices

✅ **Flow**

NED
↓
Protocol (CLI / NETCONF / RESTCONF)
↓
Network Device

🔷 **10. Network Devices (Final Execution)**

✅ **Devices**

- Routers
- Switches
- Firewalls
- Virtual devices (NetSim)

✅ **Result**

👉 Configuration applied successfully ✅
